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Exploring and building autism service capacity in rural and remote regions: Participatory action research in rural Alberta and British Columbia, Canada

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Abstract

Based in participatory action research, this project had the aim of building capacity in treatment and support for individuals and families impacted by autism spectrum disorder in remote and rural communities of Canada. Communities were selected based on their rurality and willingness to engage in change efforts for enhanced service delivery within their region. Fifteen discussion groups with key stakeholders were convened in seven communities with ~200 community stakeholders. Based on analyses of these data from the stakeholders, themes were distilled through interpretive description, which in turn were presented to community stakeholders for reflection and collective action. Findings indicate broad thematic domains consisting of: insufficient services, protective factors in community, change efforts via collectivity within community, limitations and benefits of residing in rural communities relative to care associated with autism spectrum disorder, a sense of "community" in rural contexts, and engaging in focused dialogue as a pathway to advancement. Opportunities for building capacity for support in autism spectrum disorder emerged within intersecting layers of leadership, contextual factors, and community collaboration. Consistent with participatory action research principles, emerging local knowledge was supported with strategies for improved autism spectrum disorder service development.

Keywords

autism spectrum disorder, participatory action research, rural and remote services

Introduction

We partnered with rural Canadian communities to build capacity for improved services and opportunities for people with autism spectrum disorder (ASD) and their families. This built on a community asset and partnership approach, in mutually identifying community strengths and potential steps for moving forward. Based on principles of participatory action research (PAR), the project aimed to celebrate and engage community strengths, resources, and perspectives; partnership among rural and professional communities was key to this process. In seeking ways to engage with and support rural communities, the project team invited community appraisal, stakeholder engagement, strengths and asset identification, need/priority determination, partnership building, and strategy implementation.

Background

Canada is the world's second largest country and over 90% of its land mass is considered rural (Williams and Kulig, 2012). *Rural* is a broad term that lacks a universal definition (Crosato and Leipert, 2006; Kulig et al., 2013;

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Williams and Kulig, 2012); hence, estimates of the size of the rural population in Canada vary between 19% and 30% (Statistics Canada, 2016a; Williams and Kulig, 2012). Rural and remote regions in Canada are diverse in history, ethnicity and culture, and collectively are integral to Canadian national identity (Williams and Kulig, 2012), including the social and historical contexts of survival and a pervasive sense of having to rely upon one's neighbor. Despite the history and importance of rural influences, urban centers increasingly are the focus and context of research and resource centralization; thus, arguably aggregating expertise and reducing the acknowledgment of "the crucial need for studies of dynamic Canadian rural spaces." (Williams and Kulig, 2012: 2).

The Canadian Autism Spectrum Disorder Alliance 2014 report "National Autism Needs Assessment Survey" identified the need for, "targeted outreach to address the unique issues of Canada's northern communities" (Weiss et al., 2014: 6). Also, there was a call for, "outreach to linguistically and culturally diverse communities, facilitating understanding of their service needs and improving their access to service" (Weiss et al., 2014: 6). The report identified, "(s)ervice needs in northern communities" (Weiss et al., 2014: 6) as a call for further research. In part, this call was driven by the disproportionally low number of respondents from rural, remote and northern communities (Weiss et al., 2014). The priorities outlined in the report reflected the need to understand the unique issues of rural and remote dwellers impacted by ASD, which is consistent with a broader health care literature (Williams and Kulig, 2012).

Rural health care delivery

Although distance and isolation from tertiary care centers contribute to health care access challenges in rural and remote locations, Bourke et al. (2012) emphasize the significance of structural issues for rural and remote care. Rural health does not exist in isolation; rather, it intersects with broader economic, political, and social contexts such as "macro level policies, funding, health systems, social determinants, and a range of broader processes" (Bourke et al., 2012: 198). The *Canada Health Act* governs the delivery of health services and contains principles (accessibility, universality, portability, comprehensiveness, and public administration), aiming to support "equitable access to health care for all" (Williams and Kulig, 2012: 5). Nevertheless, Canada's health funding structure contributes to a complex health care delivery system. Two decades ago, the federal government decentralized health care delivery (Zhong, 2010) and transfer payments to provinces, with decision making regarding service delivery under provincial/territorial authority (Williams and Kulig, 2012). Of further complexity, health services in Indigenous communities, which are largely represented in rural or remote regions, variably have been transferred to Indigenous health authorities. As an example of varying models, health services in Indigenous communities in British Columbia, rest with the First Nations Health Authority, with the aim of further decentralizing health care delivery and making it more culturally appropriate and responsive.

Rural health and social service care delivery for families in which an individual has a disability

Although there is limited research that specifically addresses the experiences of rural families impacted by ASD (Hoogsteen and Woodgate, 2013), there is emerging literature about rural families with children with ASD as well as a broader range of disabilities (Ashburner et al., 2016; Dew et al., 2013, 2014; Hoogsteen and Woodgate, 2013; Miller and Ward, 2013; Murphy and Ruble, 2012). Families are challenged by difficulties accessing screening, diagnostic, and interventional services because of geographic isolation, transportation limitations (e.g. inaccessibility by road, unpredictable weather conditions negatively impacting travel, lack of publicly available transit), considerable cost and time associated with traveling to obtain and/or provide services, limited numbers of health care providers responsible for widespread service delivery to a broad geographic catchment area, and incremental limits to services after transition from early childhood (Dew et al., 2013, 2014; Miller and Ward, 2013). Difficulty recruiting and retaining qualified service professionals is a frequent challenge in rural and remote communities (Kulig et al., 2013; Murphy and Ruble, 2012; Williams and Kulig, 2012).

Hoogsteen and Woodgate (2013) completed a qualitative study exploring the experience of parenting a child with ASD in a rural area. Parents experienced challenges trying to ensure that their child with ASD would experience optimal quality of life (Hoogsteen and Woodgate, 2013). Reported difficulties in rural communities included a lack of knowledge and understanding about ASD, requiring parents to educate others about ASD as well as advocate for their child (Hoogsteen and Woodgate, 2013). In addition, many families who had received a diagnosis had limited follow-up by physicians (Hoogsteen and Woodgate, 2013), consistent with reports that individuals living in rural and remote areas have limited access to specialized services (Murphy and Ruble, 2012). Given challenges and gaps in service delivery and access in rural communities, caregivers of children with disabilities often must decide whether to remain in their community or move to an urban center for better access to services (Dew et al., 2014). Overall, parenting a child with ASD in a rural community was reported to entail considerable challenges related to a lack of accessible services and resources, and a feeling of being on one's own (Hoogsteen and Woodgate, 2013).

Conversely, Hoogsteen and Woodgate (2013) while also noting parental "isolation" in relation to the restricted access or lack of access to professional services also found that support in rural communities reportedly promoted a sense of inclusion. For rural families, the social environment was noted to be an important component of well-being and reduced isolation. Dew et al. (2014) found that the social environment (friends, family, and community) contributed to social and emotional well-being for rural families in which a child had a disability.

Overall, the literature is limited and mixed in terms of the impacts of rurality relative to ASD care, and individual and family experience. Additional information is needed about the perspectives and experiences of rural dwelling for individuals with ASD and their families, and how care is navigated in this context. The aim of this project was to gain a better understanding of the issues as well as to facilitate community-based action to redress perceived inadequacies in service delivery. Project priorities were the inclusion of individuals with ASD, their families, and the broader community in determining community strengths and gaps, and identifying priorities for potential service-based and experiential advancement. Examining how these issues could be addressed meaningfully "in community" based on the priorities of the community, constituted core values of this project. To that end, we respected, relied on, and sought to reflect the perspectives of community members.

Geographic scope

The project took place in rural, remote, and northern regions of Alberta and British Columbia, Canada: provinces with large rural and remote areas and overall populations of approximately 4.2 million and 4.7 million, respectively (Statistics Canada, 2016b), living on land masses of 661,848 (Alberta) (Stamp, 2009) and 944,735 km^2 (British Columbia) (Robinson, 2010). Excluding the urban centers of greater than 500,000 inhabitants, the remaining population is sparsely distributed, generally in locations with viably harsh climates, lengthy winters with cold temperatures and abundant snow accumulation, as well as geographic barriers that limit transportation.

Williams Lake, British Columbia, and Fort McMurray, Alberta served as pilot centers, and the team also accessed communities within an approximately 3-h drive from these community hub locales. Williams Lake and Fort McMurray were chosen as they represented diversity in rural regions; the former offering a relatively stable rural community and economy with a recently established ASD center, and the latter representing a largely resource-based and changing economy, with infrastructural and ASD service gaps. Additional community sites were also selected based on relative proximity to pilot centers as well as isolation.

Williams Lake is the largest center of the Cariboo Chilcotin region of British Columbia, and services an area roughly the size of Ireland with a number of rural and remote communities, as well as 15 Indigenous communities from the Dakelh, Secwepemc and Tsilhqo'tin Nations. With a population of approximately 11,000, the community is a service hub to a region of approximately 30,000 people. It was founded on traditional Secwepemc territory during the Cariboo Gold Rush of the 1860s, but its primary industries are now forestry, logging, milling, mining, and ranching. The population has remained relatively stable over the last several decades, varying slightly with economic ebbs and flows reflecting the industrial sectors of the region. Services for children with ASD and their families are largely offered through the Cariboo Chilcotin Child Development Center, a well-established resource in the region, and the recently opened Cariboo Chilcotin Autism Center. The Child Development Center offers services to surrounding remote and rural areas, with families often coming from distant regions for assessment and support.

Fort McMurray is located within the Regional Municipality of Wood Buffalo (RMWB) in Northern Alberta, approximately 440 km north of the nearest major urban center of Edmonton. Surrounding communities in the RMWB represent First Nations, Metis, and rural/remote communities. The 2015 Municipal Census recorded an overall population of 125,032 in the region, a 143% increase from 2000 (RMWB, 2015) due to a focus on the extraction of oil from the Athabasca Oil Sands in the region, the third largest known deposit of oil in the world (Dorow and O'Shaughnessy, 2013). This rapid growth in industrial development and population has placed significant demands on social and health services to keep pace (Dorow and O'Shaughnessy, 2013). ASD-specific resources and supports are notably underdeveloped. Additional pilot communities in this project consisted of 100 Mile House, British Columbia; Tl'esqox (Toosey First Nation in British Columbia); Westlock, Alberta; Whitecourt, Alberta; and Edson, Alberta; as well as a collection of rural townships and villages in rural central Alberta. Each of these communities, as exemplified by the descriptions of Williams Lake and Fort McMurray, has unique elements associated with their respective region, but all report limited access to ASD services.

Methods

Participatory engagement

Seventeen discussion groups (five in Fort McMurray; five in Williams Lake; two in Westlock; one in Whitecourt; one in 100 Mile House; one in Tl'esqox; one in Edson; one in Central Alberta) were facilitated between March 2016 and October 2017. Prior to engagement in these discussion groups, the aims and methodology of the project were

developed, reviewed, and refined with input from community leaders and key stakeholders (e.g. service providers, educators, community groups).

Discussion groups with community stakeholders (parents, caregivers, siblings, service providers, etc.) were held in a variety of community venues, including community centers, schools, the local YMCA, golf club, hotel conference rooms, child care centers, and a Band Office The discussion groups comprised a cumulative total of approximately 200 stakeholders (across communities), typically lasted for 2–3 h per session, and ranged from 10 to 60 participants per session.

Methodological approach

PAR (Baum et al., 2006; McTaggart, 1994) consists of a mutually engaged process of inquiry between research and community team members (Minkler, 2000). What distinguishes PAR from conventional research methods is the integrally active involvement of the people whose lives are directly impacted by the issues being researched (Baum et al., 2006; Minkler, 2000). Informed by a social justice orientation that "brings together several elements of research—inquiry, learning, critical analysis, community building, and social change" (Rutman et al., 2005: 155), PAR offers a research approach consistent with the aims of this project. The project team, consisting of families, community activists, service leaders, ASD professionals, graduate students, and researchers, met in these communities to identify aims; share knowledge; elicit strengths, gaps, and needs; and engage in strategies for community capacity building. Discussion groups reflected a process of inviting stakeholders to an opportunity to dialogue about regional ASD services. For larger groups, educational content about ASD was offered prior to stakeholder discussion.

With communities, the team identified goals of broader knowledge sharing and heightened access to resources within the region, with the aim of engagement and capacity building. Based on the PAR approach, the team sought to invite, honor, and reflect priorities of local communities. Discussion groups in communities specifically addressed: (1) how individuals with ASD and their families access and navigate services, (2) what elements in the community foster, and conversely limit, resource accessibility and engagement, and (3) what is perceived as needed to improve resources and community experience. Institutional research ethics board approval was obtained prior to commencement of this research through provincial university ethics boards. Communities were central in decision-making processes throughout the research planning and implementation.

Data analysis

Initial environmental scans of the respective regions identified services, strengths, and gaps in communities, including resources and areas for development. Discussion groups complemented these findings via data which were recorded and reviewed for emerging themes. After collecting data from community stakeholders, themes were distilled through interpretive description (Thorne et al., 2004) and presented to community stakeholders who were engaged in Delphi group discussions about findings and potential pathways forward.

Following the principles of PAR, emerging knowledge was developed with significant input from participants, supported by interpretive description analysis (Hunt, 2009) of transcribed community discussions. Interpretive description seeks to describe the "smaller scale qualitative investigation of a clinical phenomenon of interest to the discipline for the purpose of capturing themes and patterns within subjective perceptions and generating an interpretive description capable of informing clinical understanding" (Thorne et al., 2004: 5). Interpretive description lends itself to developing "on-the-ground" practice recommendations, toward "the development of findings that will assist … practice" (Hunt, 2009: 1289). Thorne et al. (2004) identify the product of interpretive description as "a coherent conceptual description that taps thematic patterns and commonalities believed to characterize the phenomenon that is being studied and also accounts for the inevitable individual variations within them" (p. 4), noting that this product ought to be understandable by key stakeholders who can apply this knowledge in practice. Thorne et al. (2004) caution against detailed coding methods, such as line-by-line coding, instead advising interpretation and exploration of the data by constantly asking, "What is happening here?" (p. 7). Through these discussions, insights and strategies were discussed for improved ASD service development within respective regions.

Results

The findings indicate broad themes consisting of insufficient services, protective factors, collaboration to enact change, costs and benefits of residing in rural communities relative to ASD, "community" in rural communities, and engaging in focused dialogue as a pathway to advancement. Each of these themes is outlined below, along with illustrative quotes from the data.

Insufficient services

Participants indicated the inadequacy of ASD services in rural regions relative to the needs of affected individuals, their families, and their communities. Families identified difficulties accessing information, finding and navigating services (e.g. respite, specialized interventions, recreation), and finding support (e.g. emotional, financial), and many raised concerns about insufficient school, and youth-based community and adult-based vocational resources.

The degree of concern related to barriers varied, with some participants identifying concerns similar to those

heard province-wide. Yet, some common challenges for families affected by ASD appeared to be amplified for rural dwellers. Participants often referred to the difficult intersection of disability relative to living in a rural location, suggesting that there often is an added complexity associated with rurality related to proportionately fewer trained professionals, distance to service hubs, which was associated with less service access, inclement weather patterns or seasonal changes impeding transportation, and overall limited resource access. A participant described their sense of isolation:

> You're isolated by miles, but you're also isolated in other ways … you don't have resources outside the four walls of your living situation …

"Distance" emerged as a physical and/or geographic impediment to service access, but potentially also added a psychological burden. Long distances to services often imposed barriers on families seeking to address care needs. Service providers, particularly specialists, were generally not located in or near rural and remote communities, sometimes with key professionals visiting the community only once a month or less, if at all. Service providers, such as social workers, were often responsible for caseloads that spanned large regions, with limited time to support families given extensive time in transport and under-staffing. Also, funding structures provided to families for services in distant centers required family allocations of "therapy funding" spent on travel time and costs for service providers, resulting in a reduction of direct services to the individual with ASD. Participants identified a range of needs to improve services, as illustrated below:

> It is very difficult to get respite; my child has specific needs and you have to be able to trust the person coming into your home. And we face big challenges with conflicts of interest because qualified caregivers are limited; the people with skills for respite may work for the school as well so it's considered a conflict of interest if they were to provide respite and therefore they cannot help.

> The population is small here so there are less children with ASD, so the parent has to be expert because (the) school is not expert. It also means there is a big gap in the classrooms since the last time expertise was needed for a child with ASD, there may be a few years since the last child with ASD was in that grade.

> There [needs to] be more improvement [in services to a child with ASD].

Protective factors

Rural dwellers identified positive components and benefits unique to the rurality of their community. As an example, a "sense of community" in rural regions emerged as a supportive factor, which was defined as mutual dependence and trust. Parents in one community offered the following affirmations about their town:

> In this room [at a rural service center], there are a variety of families at different points, the wealth of knowledge in this room is amazing. Getting together and sharing knowledge, we learn.

> In our rural communities, we have a long history of supporting individuals with differing abilities.

> Seeing everyone here gives me a lot of hope (for my child and community).

In multiple instances, children or adults with ASD were known by others in the community and recognized for their contribution and inherent value to the community. One parent described her community as one of the most accepting communities she had lived in and expressed a feeling of comfort due to the community's acceptance of her adult child with ASD, which in turn reportedly reduced the "weight" she felt as a parent. She viewed her son to be safe in the community, which permitted more independence and a sense of safety and repose for this caregiver. A parent who had grown up in a rural community but had moved away described the community as "welcoming," which caused her to return as she now sought a safe, accepting, accessible community for her child with ASD.

Change efforts via collectivity within community

Rurality reportedly had a bearing on communities' ability and collective efforts toward greater resources. Substantial effort was needed for change because there were fewer potential advocates, given the smaller communities affected by ASD. Relatively few parents or other advocates per community actively sought resource change and advancement, but galvanizing these advocates through collective change efforts was viewed as productive. For instance, parents in Fort McMurray initially spoke about the need for service advancement in the region, including identifying a community dream for a local autism center. Through raised funds, momentum was built and a site was found and funded, which ultimately provided a resource library for families, parent support, and a drop-in center. Through an additional grant, a service navigation and advocacy resource was implemented.

In Williams Lake, fundraised dollars had led to the development of an autism center that included an independent living training suite and a range of community programs. Accordingly, these communities illustrated collective identification of need and service advancement based on galvanizing efforts of existing resources and commitments, and building community capacity. This

process reflected engagement with community members and supporting local leaders—even if few in numbers—in the aim of advancing, building on, and motivating localized action and change efforts.

Cost/benefits of residing in rural communities relative to ASD

Participants reflected on "what resources are needed in rural regions" versus "what is perceived to be available." While some of the core worries of urban and rural families with children with developmental disabilities may not be fundamentally different, urban dwellers were generally perceived to have more access to a broader range of services compared to rural and remote community dwellers. Family members commented,

> A lot of people enjoy this community and don't want to leave here, but we have had to have discussions about moving to a city to get services.

> Parents try so hard to be positive, but when you talk to professionals I feel everything is negative, like your dreams go down the toilet every time you go to the doctor's office.

> When I die, what happens to my child, especially if services are not available in our community?

> The connections between parents [in this community] is more like two families, you're lucky if you meet service providers through school or hearsay.

For some, rurality and remoteness imposed a perceived barrier as families envisioned services they thought would be more accessible if instead they lived in an urban center. Yet when contrasted to actual service availability in some cities, presumptions of service discrepancies periodically appeared to over-estimate urban dwellers' access to services.

Despite challenges and service gaps, a revelatory moment for participants in one discussion group emerged when a member of the research team (i.e. an outsider to the community) suggested that the "grass isn't necessarily greener on the *other* urban side," based on his observation that resources in a specific rural community in some ways offered certain resource and informal support benefits that were not available in the distant urban core, particularly due to notable strengths in that particular town, including a local autism service and a strong community/peer support base.

Participants described mixed layers of experience for individuals and families living in rural areas; they described benefits of community engagement and informal support, yet also service gaps. Parents were seen as drivers of change yet within rural areas, the number of potential parents/leaders to collectively advocate for change (e.g. a proactive response to a lack of resources) reflected fewer people due to small populations. Rurality was seen as a factor contributing to service disparities, with fewer resources and fewer people to push for change. Furthermore, more inclement weather patterns due to northern climates and significant distances to service centers contributed to a sense of isolation, as did physical/geographic impediments to timely and affordable services. This sense of isolation was exacerbated by psychological isolation reflective of a smaller and more dispersed ASD community and resource base. Yet these gaps were layered in, and mitigated by, community mindedness and relational engagement in some communities, as indicated by community members who commented as follows:

> Distance doesn't have to be something that prevents interaction.

> There are many in this community who are affected by ASD but who are not here [at a capacity building meeting] because of social isolation, physical isolation, or difficulties finding childcare or inability to get off of work?

"Community of care" in rural contexts

A strength in rural communities was frequently identified as a sense of "responsibility for one another" and not assuming that a service or resource will be offered and imported from outside the community. As a result, participants described developing ingenuity and adaptation using existing resources. Multiple examples were given of community leaders who banded together to innovatively address a need through "piecing together" services for a child or adult with ASD. A service provider commented on the collective action of the community:

> I think there is a base for something really good here, we can coordinate what already exists here. We can do what we want to do.

Participants further spoke about strengths of generosity and relationality within rural communities, as illustrated by a parent who stated, "This is one of the most accepting communities I've been in. Being out here and watching the way the community is, is comforting … comforting to know it's such an accepting community."

Engaging in focused dialogue as a pathway to advancement

Several instances emerged in which an effort toward broad-based change was advanced when supportive "others" (e.g. research/action team members), that is, allies, came alongside existing community stakeholders and existing efforts for advancement. The presence of key

others such as our PAR-based university team, appeared to positively support community priorities and teams in supporting and stimulating ideas for, and actions toward, change and partnership with key resources.

This dynamic of need, collaboration, idea generation, change effort and collective action, was repeatedly described and demonstrated among target communities. There was a reported perception of the importance of external support of local goals and values, exchange of information and resources, collective support and advancement, and as such, working together based on pressing need, emerging trust and collective aims that were based on local community priorities. Gains appeared to be built on the development of relationships.

While gains were noted during the project, they were recognized as tentative particularly if not supported by structures of policy, stable funding, and service organization within communities:

> We can figure this out, we can get the services we need to get, but we need buy in from the key service providers and government.

> We need to lean on each other; we have all been there or will be there.

> But we need to work with government for ongoing funding if service and supports are to be established and sustained.

Discussion

Project aims and outcomes included greater understanding about community features, needs and strategies. Identifying communities included consideration of demographic, geographic, and climatic factors as well as community and regional resources. Rurality emerged in the data as both a limiting factor of, and a facilitative force for, community engagement and service advancement. Participants described insufficient services in their rural communities, as well as feelings of isolation and distance from resources. However, protective factors such as a "sense of community" and relational engagement with community members were also identified by participants.

To redress noted ASD service gaps in rural communities, participants conveyed and demonstrated a range of strategies and resources largely built on relational advancement and capacity building, both within the community and with key partners outside the community (often associated with this PAR initiative). Through increased partnership with stakeholders, recommended strategies were advanced and in some cases, piloted with and by local communities. For instance, priority determination through community-based discussion groups identified ASD information and support needs. Program expansion was developed, as exemplified by a successful team grant in one community for increased family advocacy and navigational services. A Facebook ASD parent support group was initiated for rural dwellers in another community. Greater engagement in ASD services and education was realized in yet another community.

Consistent with the literature, these findings suggest that decisions about service delivery and support advancement in rural and remote regions need to reflect the need, uniqueness and diversity inherent in each community including residents, culture, and priorities (Malatzky and Bourke, 2016; Williams and Kulig, 2012). There is risk that the evolving system may not "work," particularly if a given initiative lacks "fit" relative to sensitivities of, and engagement with, a particular community (Williams and Kulig, 2012).

Although the experiences of rural and remote dwellers could not be generalized as communities were distinct the project illuminated common principles of readiness for change in the aims of seeking and moving toward improved resources for people with ASD. Oetting et al. (2001) outline a framework related to community readiness theory in rural settings which consists of five dimensions: existing efforts (programs, activities, policies), knowledge about the issue, knowledge about alternative methods or policies, leadership (appointed leaders and influential community members), funding and resources. These dimensions reportedly inform community readiness as incremental: tolerance or no awareness; denial; vague awareness; preplanning; preparation; initiation; institutionalization or stabilization; confirmation and expansion; professionalization. Although these stages are sequential, the process of community readiness is reportedly fluid (Oetting et al., 2001).

While considering a given community's stage of readiness, it is suggested that potential interventions need to fit with where communities are at and as such, communities need to be included in this process through training and strategic planning (Oetting et al., 2001). Principles of community capacity development and change are described as "incremental in phases, but nonlinear and dynamic; the change process takes time; development is essentially about developing people; development builds on existing resources; development cannot be imposed from the outside; and development is ongoing" (Kelley, 2007: 151).

In an attempt to conceptualize the process of developing rural health services, Kelley (2007) describes an explanatory metaphor of a tree: "Tree growth requires antecedent conditions such as fertile soil and healthy seed; its growth occurs slowly, and is influenced by environmental conditions such as weather and pollution; and its healthy maturity can be recognized by its dense foliage" (p. 147). Antecedent community conditions are described as "sufficient … infrastructure" (Kelley, 2007: 147), with the second phase described as a catalyst that initiates community action. The catalyst can be an event, a person (i.e. local champion), education, or funding (Kelley, 2007). The next phase of the model entails the creation of the team and involving the right people (both service providers and

community members), and the final phase is concerned with program sustainability and growth by focusing on inter-related processes of strengthening the team and engaging the community (Kelley, 2007).

Applying elements of this model, the communities involved in this project were at different phases of readiness; those moving to program initiative or advancement appeared to have reached further achievement. Yet in such cases, program sustainability had increasingly emerged as a pressing challenge requiring solid funding, policy implementation, and a scaffolding of structure to solidify the innovation. In all communities, sustainability planning of emerging services were largely needed.

These findings illustrate the benefit of engaging in strategic partnership in seeking rural ASD capacity development. In rural and remote settings, there is greater chance that a critical mass may be lacking for advocacy; a lack of key leadership risks extended delay in linking community needs with strategies to address these needs. Using tools that can offer community-to-community partnership in leveraging ideas and building synergies for collective action appears to contribute to a pathway forward.

Communities demonstrated varying methods for advancing change; however, they generally drew on a strong base of relationships both within the community and in partnership with ASD stakeholders from outside the community. Beyond face-to-face relationship development, the strategic use of the Internet may assist in connecting stakeholders who are physically distant from one another and/or as a way of building engagement among relevant service centers, communities, and so on. Better use of technology may serve to nurture strategic reach to, and among, people in rural communities.

Of potential concern here, this premise of connectivity assumes Internet access in remote communities which, in some instances, may not be the case. Accordingly, it may be important to find alternative means for connection with the aim of breaking down isolation and rendering service access. Solutions will vary across regions, populations, and cultures, yet finding creative means to engage around common aims emerged as important.

In summary, this project yielded community member engagement by working together through the processes of change in the aim of advancing community priorities. Leadership and support for community capacity building emerged as ingredients for articulating components that ultimately culminated in movement to action. Accordingly, readiness seemed to be linked with community strength such as advocates, cooperation among partners, and a collective desire for change.

Limitations

It is recognized that this project was exploratory and these findings drew on qualitative data within communities, all within a Canadian context. Generalizability of findings must be considered relative to community fit and application, with recognition for the uniqueness of communities and their existing resources and readiness. Furthermore, the PAR approach to data collection required a time commitment from participants. Some individuals or families may not have been able to participate to the extent described due to conflicting schedules and/or care requirements.

Conclusion

Notwithstanding project limitations, this initiative illustrates the potential of partnership in moving toward generative change in rural communities. To that end, ASD capacity emerged within intersecting layers of leadership, vision, and community collaboration. Partnering communities comprised dynamic and shifting systems—an inter-connection of individuals, families, agencies/services, and other stakeholders who adjusted relative to the needs, vision, and array of community resources. Finding ways to support rural communities that may lack a collective "voice" emerged as an important aim, inviting community appraisal of readiness, stakeholder engagement, strengths and asset identification, need/priority determination, partnership development, and strategy.

Further research is recommended given the emerging state of the literature in rural capacity building in ASD, including piloting initiatives of community readiness and change. Granular analysis of the mechanisms that foster capacity advancement and community development are warranted. Evaluation of this emergent framework is invited, with greater inclusion of the range of voices in rural communities including people with ASD.

Advancement in addressing gaps emerged in the aims of service access and quality of life for rural dwellers affected by ASD, their families, and communities. Yet this project also highlighted strengths in rural and remote communities relative to people with ASD and their families, and importantly nurturing a "welcoming" community.

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